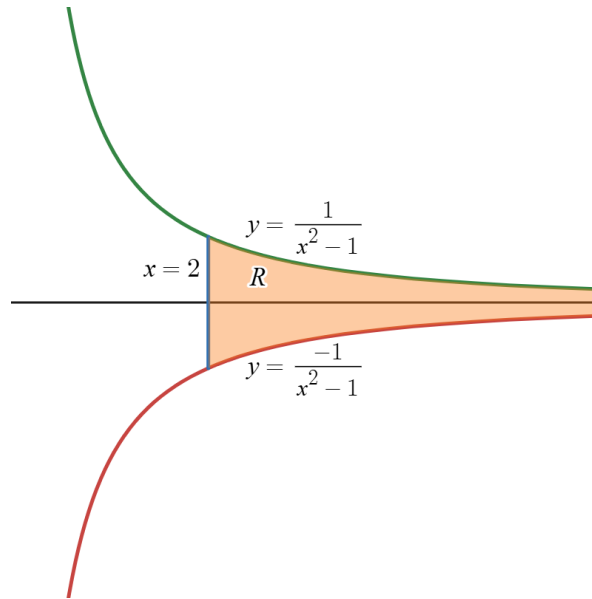


B1 past final worksheet

1. (Fall 2022, #3) Consider the infinite planar region R bounded by the vertical line $x = 2$, and the two curves $y = 1/(x^2 - 1)$ and $y = -1/(x^2 - 1)$, as shown in the graph.



- (a) Write down an improper integral that represents the area R , and decide whether it is convergent or divergent;
- (b) Calculate the area of R .

2. (Fall 2013, #4a) Let $\mathfrak{R}$ be the region of the plane bounded by the curves $y = x^3$ and $y = \sqrt{x}$. Sketch the region and find its area.

3. (Spring 2013, #3) Let A be the region under the graph of $f(x) = \frac{x^2 + x + 1}{x^2(x^2 + 1)}$, above the x -axis, and to the right of $x = 1$. Determine if A has finite area. If so, find it, and if not, explain why.

4. (Fall 2012, #4i) Consider the region $\mathcal{R}$ between the line $y = 3x$ and the curve $y = x^2 - 2x$. Sketch the region $\mathcal{R}$ and find its area.

5. (Spring 2012, #3a) Let $\mathcal{R}$ be the region bounded by the curve $y^2 = 4 - x$ and the line $x + y = 2$. Sketch the region and find its area.